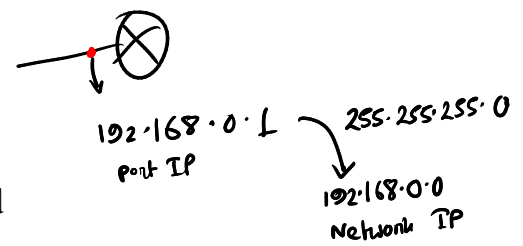


No **Static Routing** needed for directly connected networks because network addresses of directly connected networks can be figured out using subnet masks. Hence Static Routing done only for networks that arent directly connected
Does not update automatically and all changes must be applied manually



Static Routing Advantages:

- simple, predictable
- not advertised, more secure
- low CPU/Memory usage

Disadvantages:

- not scalable for large networks
- time consuming
- requires full network knowledge
- no automatic rerouting

Routing Algorithms

Internet is divided into Autonomous Systems (AS) and communication within the AS is called Intradomain Routing.

Routing Algorithms find the "good path" that has the least cost (time or financial)

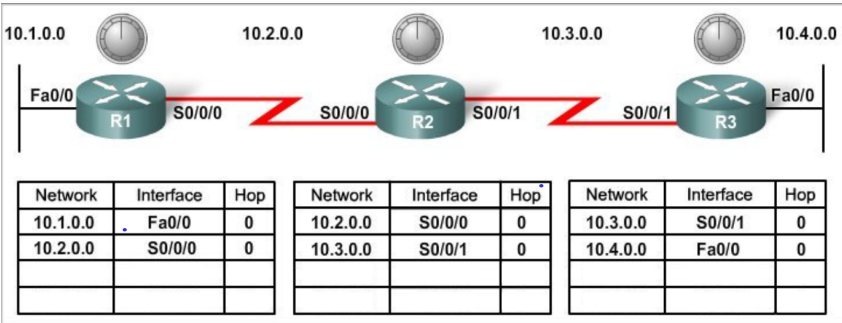
Routing algorithms are mainly :

Global: knowledge of the whole topology ex: Link State Algorithm

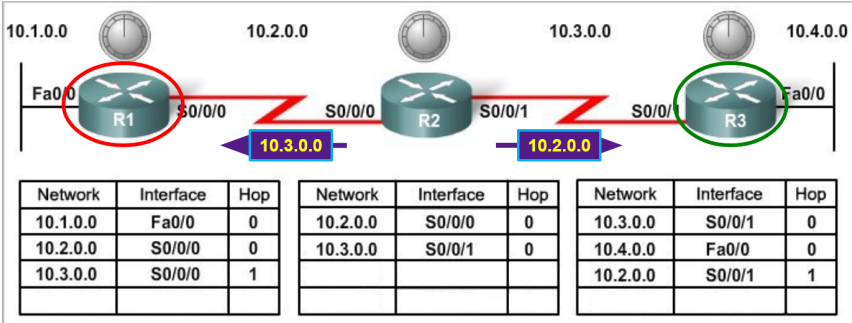
Local: knowledge of its surrounding ex: Distance Vector Algorithm

Distance Vector Algorithm

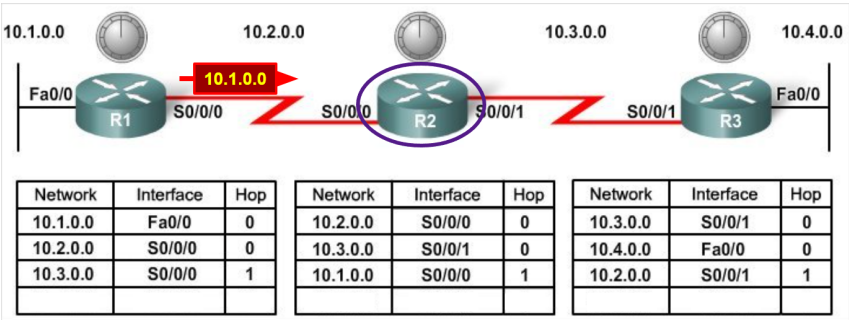
Initial
State



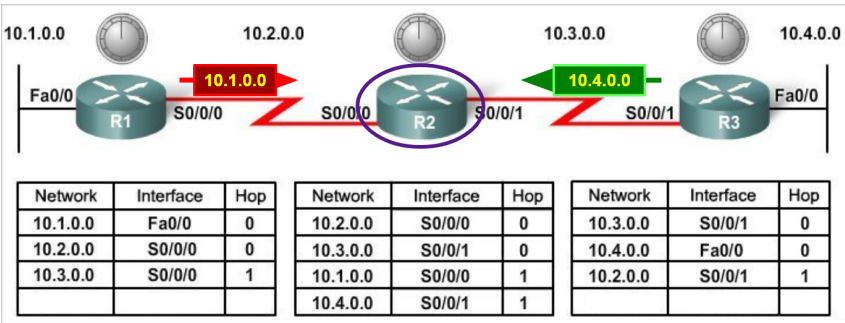
R₂ Broadcasts
to R₁ & R₃



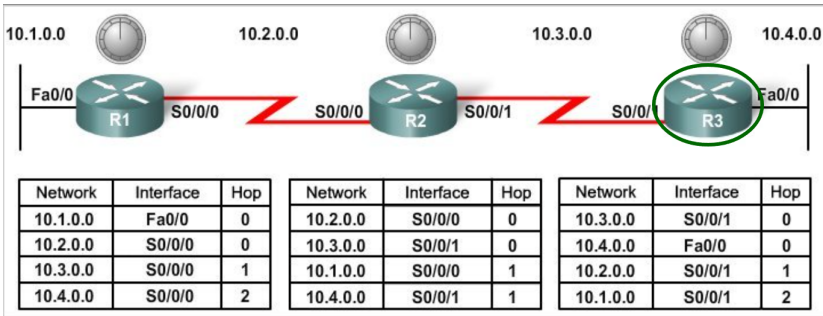
R₁ Broadcasts
to R₂



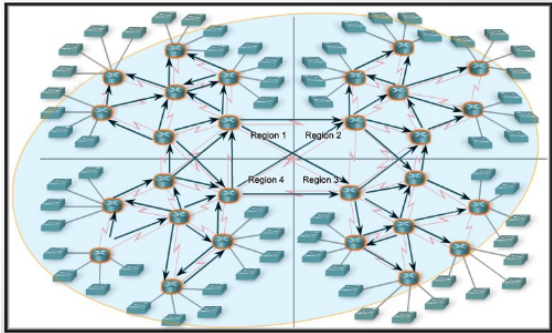
R₃ Broadcasts
to R₂



R₂ again
Broadcasts to
R₁ & R₃



Convergence is the time a network takes to reach a stable routing state after a change (e.g., link failure).



Link State Routing